

DARSINI LAKSHMIAH

DATA SCIENTIST | GENAI & APPLIED ML

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SUMMARY

Data Scientist with 4+ years of experience building AI/ML and data-intensive systems across marketing, retail, and financial services, with a focus on Generative AI, RAG, LLM evaluation, agentic applications, and production ML. Hands-on experience engineering hybrid retrieval pipelines, vector search, reranking, Dockerized REST APIs, MLflow evaluation/monitoring, and cloud-scale Python/PySpark pipelines across AWS, Azure, and Databricks. Experienced across the AI lifecycle from data preparation and retrieval design through model integration, evaluation, deployment, monitoring, and business adoption.

PROFESSIONAL EXPERIENCE

AARP | Washington, DC

Sep 2025 – Mar 2026

Data Scientist – Generative AI

- Architected a **RAG-based marketing intelligence assistant** over 1K+ campaign briefs, audience research, performance reports, and experiment readouts, enabling users to retrieve source-grounded answers with citations instead of manually searching across fragmented analytics content.
- Engineered the end-to-end **retrieval workflow** using Llama, embeddings, hybrid BM25/vector search, metadata filtering, and reranking, covering document ingestion, chunking, retrieval, context assembly, and grounded response generation.
- Productionized retrieval and LLM workflows as a **Dockerized REST API service**, packaging application dependencies and configuration and implementing authentication, structured logging, health checks, and response monitoring for reliable internal usage.
- Built a **200+ query RAG evaluation framework** in MLflow to measure Recall@5, MRR, groundedness, and citation accuracy across retrieval and generated responses, creating a repeatable quality benchmark for LLM application changes.
- Improved retrieval recall by ~15% through systematic experimentation with chunking strategies, lexical/vector retrieval, metadata constraints, and reranking, using evaluation results to identify and correct retrieval failure modes.
- Trained and operationalized a **LightGBM propensity model** across 20M+ email, web, and campaign interactions, engineering recency, frequency, channel, affinity, and response features in PySpark.
- Integrated model scoring with campaign eligibility, suppression, and audience-selection rules into reusable workflows supporting 20+ campaigns, reducing manual campaign preparation by ~30%.

Amazon | Austin, TX

May 2025 – Aug 2025

Data Scientist - Data & Analytics

- Engineered an **ML-ready customer data foundation** by cleaning and consolidating 87M+ Amazon Fresh and Whole Foods transactions from 530+ stores using SQL, Python, Spark, AWS Glue, S3, and Redshift.
- Created reusable customer features and behavioral segments from online and in-store activity, establishing a consistent feature layer for downstream customer modeling, segmentation, experimentation, and product analytics.
- Developed automated longitudinal cohort and difference-in-differences workflows to evaluate product launches across order frequency, spend, retention, and cross-banner migration, replacing repeated manual analyses with reproducible measurement pipelines.
- Designed reusable SQL/AWS data pipelines for recurring customer and product measurement, standardizing 15+ KPI definitions and improving consistency across downstream analytical and modeling workflows.

- Translated large-scale behavioral datasets into decision-ready customer signals and statistically validated findings for product leadership, connecting data engineering and modeling outputs to product adoption and customer behavior.

NatWest Group | Chennai, India

Aug 2021 – Jun 2024

Machine Learning Engineer

- Engineered **fraud-detection models** using XGBoost, TF-IDF, NLP, Python, and SQL across 1.5M+ daily card, ACH, and digital-banking transactions, reducing false positives by 12% through feature engineering and hyperparameter tuning.
- Built reusable preprocessing and feature-engineering workflows for high-volume transactional data, transforming structured transaction attributes and text-derived signals into model-ready features for fraud classification.
- Developed standardized **MLflow model inventory and monitoring workflows** for 30+ production models, tracking model versions, metadata, precision/recall, score drift, and feature stability across the model lifecycle.
- Reduced model-drift incidents by 20% through earlier identification of changes in prediction scores and feature distributions, improving visibility into production model health and supporting timely investigation.
- Built Python/PySpark and Azure Data Factory staging and integration pipelines processing 1.5M+ daily transactions for production ML systems and downstream model monitoring.
- Optimized SQL Server and Snowflake data models and queries supporting ML workloads, improving query performance by 35% and reducing data-processing bottlenecks in recurring model and analytics workflows.

PROJECTS

AI Interview Coach | AutoGen · GPT-4o · FastAPI · Whisper · WebSocket · face-api.js

- Built a real-time multi-agent interview application in which interviewer, evaluator, and candidate agents coordinate company-specific mock interviews; integrated GPT-4o, live Whisper transcription, WebSocket communication, FastAPI services, and webcam-derived performance signals.
- Designed the application around specialized agent responsibilities and real-time event exchange, combining conversational AI, speech processing, backend APIs, and multimodal signals into an end-to-end interactive AI experience.

NutriNav — Voice-First Food Access Intelligence | Groq Whisper · FastAPI · Supabase · XGBoost

- Placed Top 3 of 35 teams at George Hacks × UN Reboot the Earth; built a multilingual, voice-first AI application connecting 75,000+ DC SNAP residents with real-time corner-store inventory and USDA-based nutrition scoring.
- Integrated speech input, FastAPI services, structured inventory data, and ML-based nutrition intelligence into a deployable user workflow; judges highlighted the solution's immediate deployability and end-to-end integration.

TECHNICAL STACK

- Python, SQL, PySpark, RAG, Llama, GPT-4o, LangChain, AutoGen, Embeddings, ChromaDB, Hybrid BM25/Vector Retrieval, Reranking, Prompt Engineering, MLflow, PyTorch, TensorFlow, scikit-learn, XGBoost, LightGBM, FastAPI, Flask, REST APIs, WebSocket, Docker, AWS Bedrock, SageMaker, Lambda, S3, Glue, Redshift, Athena, Databricks, Azure Data Factory, ADLS Gen2, Snowflake, Airflow, Git/GitHub

CERTIFICATIONS

- Databricks Certified Generative AI Engineer | AWS Certified AI Practitioner | Microsoft Certified: Machine Learning Operations Engineer | Microsoft Certified: Fabric Analytics Engineer

EDUCATION

George Washington University, Washington, DC - M.S. in Data Science, GPA 3.97/4.0

May 2026

Madras Christian College, Chennai, India - B.S. in Mathematics, CGPA 8.50/10

May 2021